

Citation-Network Cohesion and Reciprocal Citation Cliques Across Journal-Impact Groups: A Matched Author Analysis

Panagiotis-Alexios Spanakis, Grigorios Alexandrou, and Diomidis Spinellis

Department of Management Science and Technology, Athens University of
Economics and Business (AUEB), Athens, Greece

Abstract

We ask whether authors in lower-impact journal groups have different citation patterns from matched authors in higher-impact groups. We call the lower-impact observations Cases and the higher-impact observations Controls. We compare 9,431 author–subject pairs within subject and at similar recent citation productivity. Cases had higher average values on all four primary measures, although three median differences were zero because many pairs tied at zero. A defined clique analysis, in which every member pair is linked and many links are returned, found 17 reciprocal cliques among 50 candidate groups; randomized graphs averaged 0.73 reciprocal cliques (empirical $p = 0.0020$), and 91.6% of clique memberships were Cases (label-swap $p = 0.0020$). The results support a directional structural association between journal-impact group and citation-network organization.

Keywords: scientometrics; citation networks; matched design; network cohesion; journal impact; anomaly screening

1 Introduction

Citation indicators are widely used to summarize scholarly influence, allocate resources, and support evaluation. Their convenience also creates incentives to optimize what is measured, which can weaken the connection between an indicator and the quality it is intended to represent (Biagioli et al., 2019; Goodhart, 1984; Hicks et al., 2015). Excessive self-citation, coercive citation, and concentrated citation exchange are therefore important objects of measurement in their own right (Biagioli & Lippman, 2020; Van Noorden & Chawla, 2019; Wilhite & Fong, 2012).

Most evidence on unusual citation structure is either journal-level or begins with groups already selected for unusual behavior. Journal-network studies can identify exceptional exchange patterns (Kojaku et al., 2021), and analyses of citation stacking show how journal-level citation flows can be distorted (Heneberg, 2016). Author-level concentration indicators can show when a small set of collaborators accounts for much of an author’s citation impact (Evdaimon, Ioannidis, et al., 2024). Less is known about whether simple author-network measures differ across journal-impact groups after authors are compared within subject and at similar levels of recent citation productivity.

This study addresses that gap with a matched author–subject design. Low-impact and high-impact refer only to groups defined by subject-specific, publication-weighted 25th and 75th percentile cutoffs of journal Eigenfactor scores. We use *Case* and *Control* as compact labels for those operational groups; neither label evaluates the scientific merit of an author or work.

The study asks three questions:

1. Do matched Case and Control authors differ in citation-network cohesion and outgoing citation concentration?
2. Are author–subject profiles that are unusual relative to Controls enriched among Cases?
3. How do fractional citation weights mix across the two groups?

We expect Case author–subjects to have higher values than Controls on four primary measures: citations to previous collaborators, reciprocal citation, local neighbor connections, and concentration of outgoing citations. This is an associational hypothesis, not a causal or integrity claim. Because many values are zero, a zero median shift does not rule out a directional difference. Network closure can reflect legitimate specialization and collaboration, 45 so the anomaly procedure is a screening device.

2 Related Work

Research evaluation based on citation counts, the h -index, and journal indicators has long been accompanied by cautions about field differences and strategic responses (Garfield, 2006; Hicks et al., 2015; Hirsch, 2005). Self-citation is common and can be substantively appropriate, 50 but extreme rates complicate comparisons (Aksnes, 2003; Van Noorden & Chawla, 2019). Coercive citation and metric gaming demonstrate that observed references can also respond to editorial and institutional incentives (Baccini et al., 2019; Wilhite & Fong, 2012).

Several strands of work motivate a network perspective. Kojaku et al. identify anomalous groups in journal citation networks by comparing observed patterns with network expectations 55 (Kojaku et al., 2021). Heneberg examines the transition from journal self-citation to citation stacking through journal-level citation flows (Heneberg, 2016). These approaches establish the value of relational structure but operate at the venue level. At the author level, Evdaimon et al. propose indicators based on the concentration of citing authors and hyper-collaboration (Evdaimon, Ioannidis, et al., 2024). Their concentration measures identify a narrow citing 60 base; local closure, reciprocity, and concentration of an author’s own references provide complementary information about how that base is connected.

The present analysis combines this network perspective with within-subject author matching. It does not attempt to classify behavior as legitimate or illegitimate. Instead, it estimates matched differences in a small, declared set of network measures and applies a 65

Control-referenced screen whose output remains explicitly descriptive.

3 Data and Cohort

3.1 Source data and study window

The source is a fixed Crossref snapshot processed with *alexandria3k* (Spinellis, 2023). The
70 study window covers citing and cited works dated 2020–2024. Available fields include
Digital Object Identifiers (DOIs), publication year, normalized journal International Standard
Serial Numbers (ISSNs), work references, and author rows with optional ORCID researcher
identifiers. ORCID is a persistent identifier for a researcher. The estimates apply to this
fixed snapshot and to the author–subject observations that meet the stated inclusion criteria.

75 Journals are assigned to Health Sciences, Life Sciences, Physical Sciences, Social Sciences
and Humanities, or Multidisciplinary. The existing classification uses a deterministic, cached
large-language-model annotation of representative journal content.

The retained venue classification uses fixed subject-specific journal scores from a separate
directed journal-citation calculation (Bergstrom et al., 2008). An Eigenfactor score is a
80 journal influence score based on citation flows. In plain terms, the calculation passes influence
from citing journals to cited journals, excludes journal self-citations, and uses article counts to
redistribute influence from journals without outgoing citations and to restart the calculation.
These journal-level scores define the two venue groups; they are an input to the comparison,
not an author-network outcome. The exact matrix specification is given in Appendix A.

85 The current workflow consumes the fixed database scores as an upstream exposure rather
than re-estimating them. After scores are attached to eligible 2020–2024 work rows, the
25th and 75th percentile cutoffs are calculated across positive-score work rows within each
subject. A journal therefore contributes once per eligible work: the cutoffs are publication
weighted, not percentiles over distinct journals. Journals at or below the lower cutoff form
90 the low-impact group and those at or above the upper cutoff form the high-impact group.

No new venue classification was undertaken for this revision.

3.2 Author–subject cohort

The unit used throughout is an *author–subject*, meaning one researcher considered within one subject area, identified by (ORCID, s). An author is a Case in subject s when at least 70% of eligible works in that subject appear in low-impact journals, and a Control when at least 70% appear in high-impact journals. Authors not meeting either rule are excluded from matching. A uniqueness check prevents more than one group assignment for the same (ORCID, s) key. 95

The retained primary cohort contains 9,431 matched Case–Control pairs across the five subjects. The deterministic 10% ORCID sample and the original matching cohort are retained to separate analytical corrections from cohort redefinition. 100

4 Matching

Recent subject-specific citation productivity is summarized by h_5 , the Hirsch index calculated from an author’s works in subject s published during 2020–2024 (Hirsch, 2005). In ordinary language, h_5 is the largest number of works that each have at least that many citations in the fixed snapshot. The retained upstream table contains only positive values, so this fixed-cohort 105 revision restricts the matching pool to $h_5 \geq 1$ and excludes otherwise group-eligible zero- h_5 profiles before candidate generation.

We pair lower-impact and higher-impact author–subjects within the same subject when their h_5 values differ by at most three points. Candidates are considered in the fixed order subject, smallest absolute h_5 difference, Case ORCID, and Control ORCID; each 110 author–subject can be used only once.

An independent audit rebuilds this process from the source profiles and compares it with the retained cohort. It reproduced the retained pair membership exactly within the positive- h_5 pool; no pair exceeded the three-point limit and no author–subject key was reused.

Table 1: Matched authors and balance by subject.

Subject	Pairs	h_5 observed	Exact h_5	Median Δh_5	Median $ \Delta h_5 $	Maximum $ \Delta h_5 $
Overall	9,431	9,431	5,736	0.00	0.00	2
1	3,776	3,776	1,651	-1.00	1.00	2
2	2,543	2,543	1,319	0.00	0.00	2
3	332	332	290	0.00	0.00	1
4	2,262	2,262	1,958	0.00	0.00	1
5	518	518	518	0.00	0.00	0

Δh_5 is Case minus Control. We matched only pairs with an absolute difference of at most three. Exact- h_5 pairs are a sensitivity check, not a replacement for the main cohort.

115 The complete audit accompanies the replication materials.

The matching table supplies the Case ORCID, Control ORCID, subject, and both h_5 values. Feature rows are linked back to this table twice, on $(\text{case_orcid}, s)$ and $(\text{control_orcid}, s)$; ORCID-only joins are not used.

We report the distribution of h_5 and the within-pair difference before outcome testing.
 120 Because residual differences can remain under caliper matching, the four primary comparisons are repeated on the subset with exact h_5 equality.

The generated cohort audit found a maximum observed absolute h_5 difference of 2 and 0 pairs beyond the three-point caliper.

5 Network Construction and Definitions

125 We represent authors as nodes and citations as links between nodes. A link has a direction, from the citing author to the cited author, and a fractional weight that accounts for the number of authors on both works. We use the word “network” for this set of nodes and weighted links.

5.1 Reference resolution and fractional weights

130 A citing-work/cited-work DOI pair is deduplicated before either work is expanded to author rows. Let n_c and n_a be the complete counts of author rows on the citing and cited work.

These denominators include authors without ORCIDs. A resolved work citation contributes

$$\frac{1}{n_c n_a} \tag{1}$$

to each identified citing-author/identified cited-author combination. Rows with an unidentified endpoint do not produce analytic author edges; in particular, missing identifiers cannot become a shared pseudo-author. Multiple matched authors on one citing work do not multiply the 135 underlying resolved work reference.

Every analytic edge is keyed by subject, citing ORCID, cited ORCID, and citation year. Only citing works assigned to the matched author’s subject contribute to that author–subject’s measures. Duplicate rows are aggregated so that this key is unique.

For coauthor classification, let c_{ij} be the first year in which authors i and j published 140 together. A citation from i to j in year t is a prior-coauthor citation only if $c_{ij} < t$. A sensitivity analysis uses $c_{ij} \leq t$ to show the effect of admitting same-year collaborations.

5.2 Graph and reference populations

The analysis uses three views of the citation links so that each measure has a clear population.

Each author’s outgoing links. For each matched author–subject i , this view contains 145 citations from i to any identified author reached from an eligible citing work. It defines self-citation, citations to previous collaborators, outgoing concentration, and the largest year-to-year increase to one recipient.

Links among matched authors. For each subject s , this view retains only matched authors and the links between them. It defines reciprocal citation, local neighbor connec-150 tions, the balance of incoming and outgoing weight, and citation mixing between the two groups.

Deduplicated work references. This view is built before author expansion and defines

same-journal referencing even when the cited author is not identified.

155 Keeping these views separate prevents a measure based on all identified recipients or work
 references from being mistaken for a measure based only on matched authors.

5.3 Notation, metrics, and missing values

Some measures follow an author’s outgoing citations to any identified recipient; others use
 only links between matched authors. Direction is ignored only for the simple neighbor graph
 160 used for local clustering.

Let x_{ijt} denote the fractional citation weight from author i to author j in year t , after the
 construction above. Missing dyad-years are zero, and $w_{ij} = \sum_{t=2020}^{2024} x_{ijt}$ is cumulative weight.
 If V_s is the set of matched authors in subject s , the matched-author graph is the directed,
 weighted, loopless graph

$$G_s = (V_s, E_s, w), \quad E_s = \{(i, j) : i, j \in V_s, i \neq j, w_{ij} > 0\}. \quad (2)$$

165 Its simple binary undirected projection is

$$U_s = (V_s, \tilde{E}_s), \quad \{i, j\} \in \tilde{E}_s \iff i \neq j \text{ and } w_{ij} + w_{ji} > 0. \quad (3)$$

Direction and fractional weight are discarded only in U_s . Self-edges remain available in the
 outgoing ego layer for self-citation and surge share but do not enter G_s or its peer-network
 measures.

5.4 Formal reciprocal-clique analysis

170 We test citation-clique structure on all matched authors, not only on authors flagged by
 the screen. A clique C is a group of at least four matched authors in which every pair is
 connected by a citation in at least one direction. We keep only maximal groups, meaning that

no additional matched author can be added without breaking that rule. The direction and weight of the links are then evaluated separately. For a clique of size k , directed density is

$$d_C = \frac{m_C}{k(k-1)}, \quad (4)$$

where m_C is the number of positive directed dyads among the k members. Weighted reciprocity is

$$r_C = \frac{\sum_{\{i,j\} \subset C} \min(w_{ij}, w_{ji})}{\sum_{\{i,j\} \subset C} \max(w_{ij}, w_{ji})}. \quad (5)$$

The primary rule requires at least 75% of all possible directed links and a weighted reciprocity of at least 0.50. We repeat it for minimum clique sizes $k = 3, 4, 5$ and reciprocity thresholds 0.25, 0.50, 0.75; the density threshold remains fixed at 0.75. The anomaly flag is reported for these groups but does not define them. 180

To ask whether the observed count is unusual, we create randomized versions of each subject network. Each random version keeps every author’s number of incoming and outgoing links, but changes which authors are linked. We then score the same candidate cliques with the same rules. We use 500 independent random versions and report the proportion that reach or exceed the observed value, with the standard add-one correction. A separate set of 500 runs swaps Case and Control labels within each matched pair; this tests whether clique membership is concentrated in Cases. These comparisons test whether the defined pattern is unusual and where it is concentrated. They do not explain why it occurred.

The study-specific temporal measure is

$$B_i = \frac{\max_{j,t=2021,\dots,2024} [x_{ijt} - x_{ij,t-1}]_+}{\sum_j \sum_{t=2020}^{2024} x_{ijt}}. \quad (6)$$

This study-specific temporal increment is not a state-transition burst-detection model. 190

Unavailable values remain missing through feature assembly. A zero is used only when the denominator and graph population are observed but the relevant event is absent. Thus a

Table 2: Declared author–subject measures.

Measure	Definition	Population and interpretation
Prior-coauthor citation rate	$\frac{\sum_{t,j \neq i} x_{ijt} \mathbf{1}(c_{ij} < t)}{\sum_{t,j \neq i} x_{ijt}}$	Outgoing ego layer. Fraction of identified non-self outgoing weight directed to an author whose first collaboration preceded the citation year.
Self-citation rate	$\frac{\sum_t x_{iit}}{\sum_{t,j} x_{ijt}}$	Outgoing ego layer. Fraction of identified outgoing weight assigned to the same ORCID.
Reciprocity	$\frac{\sum_{j \neq i} \min(w_{ij}, w_{ji})}{\sum_{j \neq i} w_{ij}}$	Directed, weighted G_s . Among matched peers, the share of outgoing weight that is also returned to the author.
Local clustering	$\frac{2T_i}{k_i(k_i - 1)}$	Binary projection U_s . Among an author’s matched neighbors, the share of neighbor pairs that are connected to each other. Here k_i is the number of matched neighbors and T_i is the number of connected pairs among those neighbors.
Outgoing Herfindahl–Hirschman index (HHI)	$\sum_{j \neq i} p_{ij}^2, p_{ij} = w_{ij} / \sum_{k \neq i} w_{ik}$	Outgoing ego layer. Concentration of identified non-self outgoing weight across recipients; values near 1 mean that one recipient receives nearly all of the weight (Hirschman, 1964).
Journal endogamy	Resolved outgoing work references whose citing and cited journals have the same nonblank normalized ISSN, divided by all resolved outgoing work references	Deduplicated work-reference layer. Share of resolved references whose citing and cited journals have the same normalized ISSN. This descriptive term does not imply impropriety.
Within-sample citation balance	$\frac{\sum_{j \neq i} w_{ij} - \sum_{j \neq i} w_{ji}}{\sum_{j \neq i} w_{ij} + \sum_{j \neq i} w_{ji}}$	Directed, weighted G_s . Positive values mean more outgoing than incoming weight among matched peers; negative values mean more incoming weight.
Maximum annual dyadic surge share	$\frac{\max_{t=2021, \dots, 2024} \sum_j [x_{ijt} - x_{ij,t-1}]_+}{\sum_j \sum_{t=2020}^{2024} x_{ijt}}$	Outgoing ego layer. Largest year-to-year increase to one recipient, divided by all identified outgoing weight.

Notes. x_{ijt} is fractional citation weight, w_{ij} is cumulative weight, c_{ij} is the first joint-publication year, and $\mathbf{1}(\cdot)$ is an indicator. G_s is the directed matched-author graph, U_s is its binary neighbor graph, and $[z]_+ = \max(z, 0)$. Measures are calculated separately for each author–subject; the zero and missing-value rules follow the table.

defined rate with no qualifying citations is zero; reciprocity is zero when internal outgoing weight exists but none is returned; clustering is zero for an observed induced-graph node with fewer than two neighbors; and journal endogamy is zero when resolved references exist 195 but none remains within the citing journal. Outgoing HHI is missing when there is no identified non-self outgoing weight, balance is missing when total internal incident weight is zero, journal endogamy is missing when no resolved outgoing work reference exists, and surge share is missing when total identified outgoing weight is zero. These rules imply $0 \leq B_i \leq 1$. Statistical comparisons use complete-pair deletion separately for each metric rather than a 200 common imputed matrix.

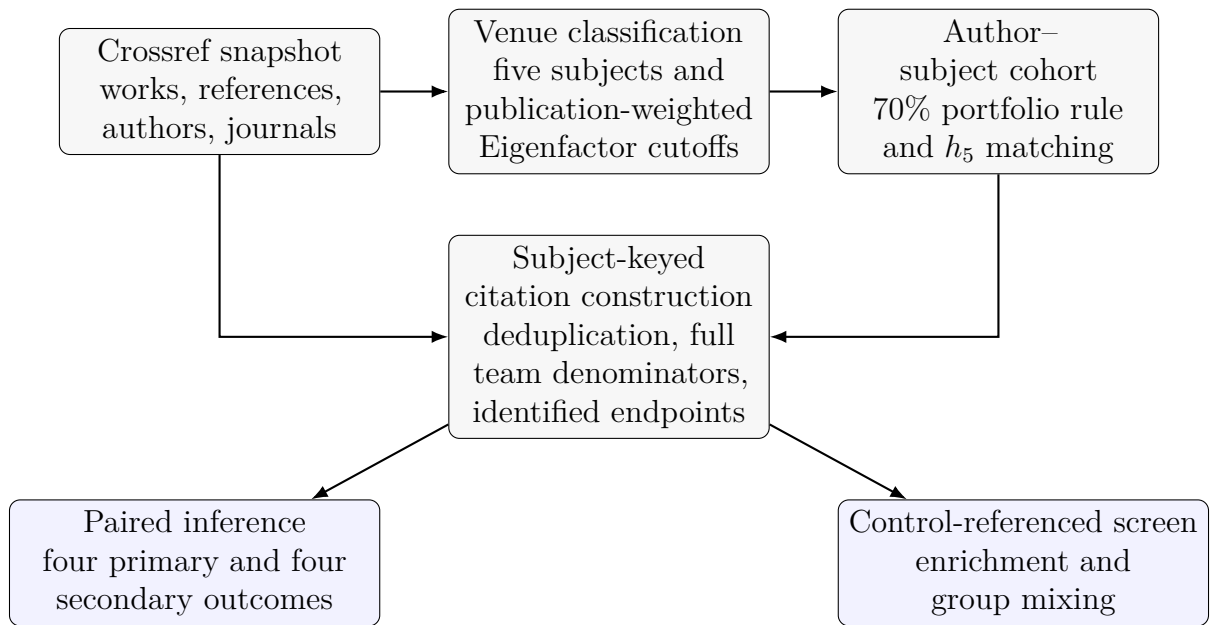


Figure 1: Reader-facing study design. Venue classification and author matching define the cohort; subject-keyed citation construction then supports two declared analytical branches.

6 Paired Inference

The four primary outcomes are citations to previous collaborators, reciprocal citation, connections among an author’s citation neighbors, and outgoing HHI. HHI is a concentration score from 0 to 1; it is higher when an author sends most outgoing citation weight to a small number of recipients. The four secondary outcomes are self-citation, same-journal referencing, the balance of incoming and outgoing links, and the largest year-to-year increase to one recipient. The former normalised burst-intensity feature is not an additional outcome; it was replaced by the surge share in Equation 6. The revised analysis therefore reports eight distinct outcomes in two declared families. 210

For each matched pair, we subtract the Control value from the Case value. A positive difference therefore means that the Case value is larger. Only pairs with both values available enter a comparison. Because the measures contain many ties and zeros, we use the two-sided paired Wilcoxon signed-rank test, which does not assume a normal distribution (Wilcoxon, 1945). We report the number of pairs, the median difference, and a rank effect size: 215

$$r_{\text{rb}} = \frac{W^+ - W^-}{W^+ + W^-}, \quad (7)$$

where W^+ and W^- add the ranks of positive and negative differences after zero differences are omitted. The effect ranges from -1 to 1 ; a positive value means that Case ranks tend to be higher. We obtain 95% intervals from 2,000 bootstrap samples of matched pairs (Efron & Tibshirani, 1993). We also report mean differences and their bootstrap intervals because they give an intuitive measure of the size of the shift. Benjamini–Hochberg correction adjusts the 220 four p -values in each outcome family for the number of tests (Benjamini & Hochberg, 1995).

As a randomization check, we randomly reverse the sign of each pair difference in 10,000 runs and compare the resulting mean differences with the observed one. This keeps the size of every pair difference while removing its direction. We also repeat the primary comparisons for exact- h_5 matches and for a definition that counts collaborations formed in the citation 225

year.

7 Anomaly Screen

The anomaly analysis asks a different question from the matched comparison: are unusual profiles more common among Cases than among Controls? It is a screening rule, not a
230 prediction model and not evidence of intent.

An author–subject enters the screen only when it has at least one identified non-self outgoing citation. The screen uses five measures of an author’s citation links: self-reference, citations to previous collaborators, mutual exchange, connections among neighbors, and concentration among recipients. Same-journal referencing, the balance of incoming and
235 outgoing links, and year-to-year increases are reported separately and are not screen inputs. All five inputs must be observed; an unavailable value is not changed to zero. Controls define the subject-specific reference values, and Cases are compared with those values.

An Isolation Forest is a tree-based method that gives a higher score to a profile that is easy to separate from typical Controls (F. T. Liu et al., 2008). A row passes this part of the
240 screen when its score is above the 99th percentile of Controls in the same subject. A row also passes the second part when at least two of four cohesion measures exceed their Control percentiles. The final flag requires both parts. Because the four cohesion measures also enter the first part, the two parts are related and are not independent evidence.

Sensitivity analyses vary the Control percentile, random seed, and detector feature set.
245 We report overlap between the detector and cohesion restriction, their rank association, and leave-one-feature-out stability. These quantities describe an operational screen; they are not measures of classification accuracy because group is not a ground-truth anomaly label.

Finally, we summarize where citation weight goes. The 2×2 table has the citing group in the rows and the cited group in the columns. Weighted assortativity compares the observed
250 share of same-group citations with the share expected from the row and column totals

(Newman, 2002):

$$r = \frac{\text{tr}(e) - a^\top b}{1 - a^\top b}. \quad (8)$$

Here e is the observed 2×2 table of citation-weight shares, a contains its row totals, b contains its column totals, and $\text{tr}(e)$ is the observed same-group share. Positive r indicates more same-group citation weight than expected from those totals. The reported label-swap p -value compares the observed same-group share with 10,000 runs in which the two labels are swapped within each matched pair. The swaps keep the citation links and matching structure unchanged while removing the original group labels.

8 Results

All numerical values in this section come from the same analysis and are shown in the tables and figures.

260

8.1 Matched differences in network measures

The primary and secondary matched comparisons are shown in Tables 3 and 4.

Across the matched pairs, Cases had higher average values on all four primary measures. Three median differences were zero because many pairs tied at zero; among non-tied pairs, Case values were more often higher. The result is a shift across the paired sample rather than a larger value for every author.

Table 3: Matched comparisons of the four primary measures.

Metric	Pairs	Case median	Control median	Median diff.	Boot. CI	Mean diff.	Mean boot. CI	Rank effect	Adj. <i>p</i>	Sign-flip <i>p</i>
Coauthor-citation rate	6,848	0.000	0.029	0.000	[0.000, 0.000]	0.045	[0.039, 0.050]	0.116	< 0.001	< 0.001
Reciprocity	464	0.000	0.000	0.000	[0.000, 0.000]	0.129	[0.089, 0.170]	0.468	< 0.001	< 0.001
Local clustering	9,431	0.000	0.000	0.000	[0.000, 0.000]	0.026	[0.021, 0.031]	0.395	< 0.001	< 0.001
Outgoing HHI	6,848	0.073	0.030	0.035	[0.033, 0.038]	0.097	[0.091, 0.102]	0.580	< 0.001	< 0.001

The Pairs column gives the number of matched pairs with both values present; this count can differ by metric. Each confidence interval (CI) is a 95% range from the paired bootstrap. The rank effect is the rank-biserial effect size. Median differences can be zero because many values are tied at zero, while mean differences show the average shift. Positive differences and positive rank-biserial effects indicate Case > Control. BH adjusts the four displayed *p*-values for multiple testing; the sign-flip *p*-value comes from randomly reversing pair directions.

Table 4: Matched comparisons of secondary measures.

Metric	Pairs	Case median	Control median	Median diff.	Boot. CI	Mean diff.	Mean boot. CI	Rank effect	Adj. <i>p</i>	Sign-flip <i>p</i>
Self-citation rate	6,908	0.000	0.007	0.000	[0.000, 0.000]	0.029	[0.026, 0.033]	0.163	< 0.001	< 0.001
Journal endogamy	7,296	0.007	0.065	-0.028	[-0.030, -0.026]	-0.010	[-0.015, -0.005]	-0.244	< 0.001	< 0.001
Within-sample citation balance	1,075	0.014	-0.259	0.000	[0.000, 0.000]	0.101	[0.030, 0.177]	0.105	0.008	0.007
Maximum annual dyadic surge share	6,908	0.121	0.068	0.047	[0.044, 0.051]	0.092	[0.087, 0.097]	0.514	< 0.001	< 0.001

The Pairs column gives the number of matched pairs with both values present; this count can differ by metric. Each confidence interval (CI) is a 95% range from the paired bootstrap. The rank effect is the rank-biserial effect size. Median differences can be zero because many values are tied at zero, while mean differences show the average shift. Positive differences and positive rank-biserial effects indicate Case > Control. BH adjusts the four displayed *p*-values for multiple testing; the sign-flip *p*-value comes from randomly reversing pair directions.

Cases had higher average values on all four primary measures (Coauthor-citation rate 0.045, 95% bootstrap CI [0.039, 0.050]; Reciprocity 0.129, 95% bootstrap CI [0.089, 0.170]; Local clustering 0.026, 95% bootstrap CI [0.021, 0.031]; Outgoing HHI 0.097, 95% bootstrap
270 CI [0.091, 0.102]). Three median differences were zero because many pairs had the same zero value. Among non-tied pairs, Case values were more often higher. Outgoing HHI also had a positive median paired difference 0.035, 95% bootstrap CI [0.033, 0.038].

Cases had less same-journal referencing (median difference -0.028, 95% CI [-0.030, -0.026]) and larger year-to-year increases to one recipient (median difference 0.047, 95% CI [0.044,
275 0.051]).

Table 5: Checks using exact- h_5 matches and same-year collaborations.

Metric	Pairs	Case median	Control median	Median diff.	Boot. CI	Mean diff.	Mean boot. CI	Rank effect	Adj. p	Sign-flip p
Coauthor-citation rate	4,221	0.030	0.026	0.000	[0.000, 0.000]	0.069	[0.062, 0.077]	0.273	< 0.001	< 0.001
Reciprocity	297	0.000	0.000	0.000	[0.000, 0.000]	0.167	[0.114, 0.220]	0.511	< 0.001	< 0.001
Local clustering	5,736	0.000	0.000	0.000	[0.000, 0.000]	0.036	[0.031, 0.042]	0.556	< 0.001	< 0.001
Outgoing HHI	4,221	0.071	0.030	0.034	[0.031, 0.037]	0.097	[0.090, 0.104]	0.569	< 0.001	< 0.001
Coauthor-citation rate (same-year included)	6,848	0.066	0.081	0.000	[-0.003, 0.000]	0.052	[0.045, 0.058]	0.102	< 0.001	< 0.001

The Pairs column gives the number of matched pairs with both values present; this count can differ by metric. Each confidence interval (CI) is a 95% range from the paired bootstrap. The rank effect is the rank-biserial effect size. The first four rows use exact- h_5 matches and BH correction across the primary outcomes. The same-year coauthor row uses the full cohort as a separate one-outcome check. Median differences can be zero because many values are tied at zero, while mean differences show the average shift.

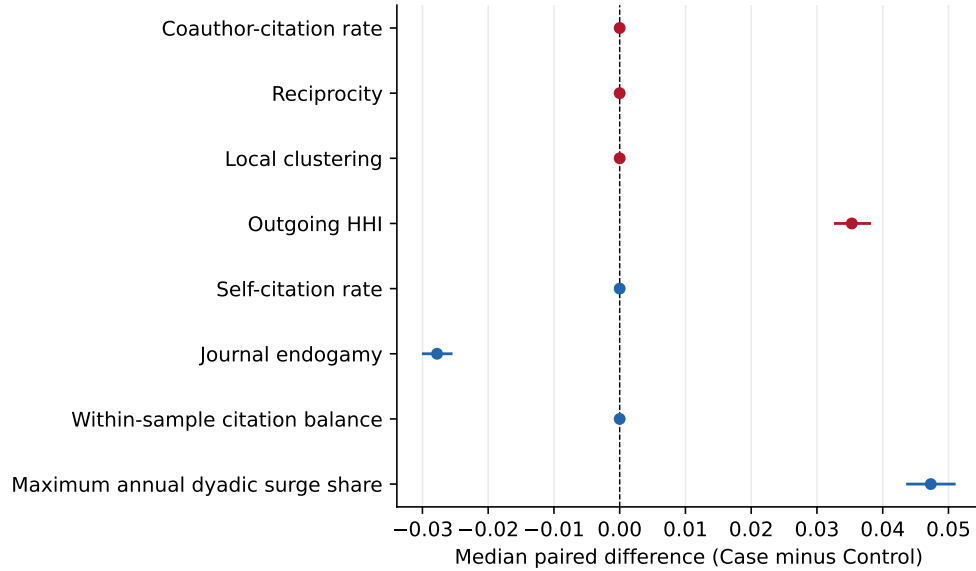


Figure 2: Matched effect estimates by outcome. Points show median paired differences and bars show paired-bootstrap 95% confidence intervals; mean differences and mean intervals are reported in Tables 3 and 4.

Pair counts are reported separately because the missingness pattern differs by measure. The count for every metric is checked not to exceed the 9,431 primary pairs.

Table 6: Screen results by journal-impact group.

Tier	Eligible	Screenable	Flagged	Flagged share	Case/Control enrichment	Case share of flags	Fisher exact p
Case	6,960	1,897	49	0.70%	16.14	92.45%	< 0.001
Control	9,171	1,973	4	0.04%	16.14	–	< 0.001

Eligibility requires identified non-self outgoing citation weight. Enrichment is the Case flagged share divided by the Control flagged share. The Fisher exact p -value tests whether the two flagged shares differ. Flags are screening results and do not establish intent or misconduct.

Table 7: The two screen rules and their overlap.

Detector exceeds threshold	Cohesion rule: no	Cohesion rule: yes	Total
No	3,668 (94.78%)	2 (0.05%)	3,670 (94.83%)
Yes	147 (3.80%)	53 (1.37%)	200 (5.17%)
Total	3,815 (98.58%)	55 (1.42%)	3,870 (100.00%)

Cells are counts among 3,870 screenable author–subject rows. The detector rule uses the tree-based score; the cohesion rule requires at least two of four measures to exceed their Control references. The two rules share four measures, so their overlap is not independent evidence. Their scores were positively related ($\rho = 0.385$, $p < 0.001$).

8.2 Sensitivity of primary comparisons

The exact- h_5 check kept the direction of all four primary comparisons; outgoing HHI had a
280 median difference of 0.034 (95% CI [0.031, 0.037]). Using the same-year coauthor definition, the median difference was 0.000 (95% CI [-0.003, 0.000]).

The check keeps the direction of the primary results. The table shows the estimates and intervals for the alternative matching and collaboration definitions.

8.3 Control-referenced anomaly enrichment

285 The screen flagged 49 Cases and 4 Controls (0.70% versus 0.04%; 16.14-fold difference; $p < 0.001$).

The two screen rules agreed for 53 rows; 147 passed only the first rule and 2 only the second. Their scores were positively related ($\rho = 0.385$, $p < 0.001$).

Removing one input at a time changed little: flag overlap ranged from 0.907 to 0.981,
290 and Case/Control enrichment ranged from 15.15 to 16.80. Details appear in Table 12.

Table 8: Sensitivity of screen results.

Control percentile	Flagged count: median [range]	Case share	Control share	Enrichment	Overlap with main screen
97.5%	97 [93–99]	1.21%	0.13%	9.22	0.55
99.0%	54 [52–55]	0.71%	0.04%	16.31	0.98
99.5%	42 [41–44]	0.55%	0.04%	12.68	0.77

Each row summarises ten fixed seeds; the main screen uses the 99th percentile and seed 42. Enrichment is the Case flagged share divided by the Control flagged share. Overlap is the Jaccard share of flagged rows retained by both screens.

Table 9: Where matched authors cite.

Citing group	Cited Case	Cited Control
Case	226.87	14.78
Control	3.73	58.38

Cells are cumulative fractional citation weights. The within-group share is 93.91%, and weighted assortativity is 0.824; this compares same-group weight with the amount expected from the row and column totals. The two-sided 10,000 within-pair group-label-swap p-value is < 0.001 .

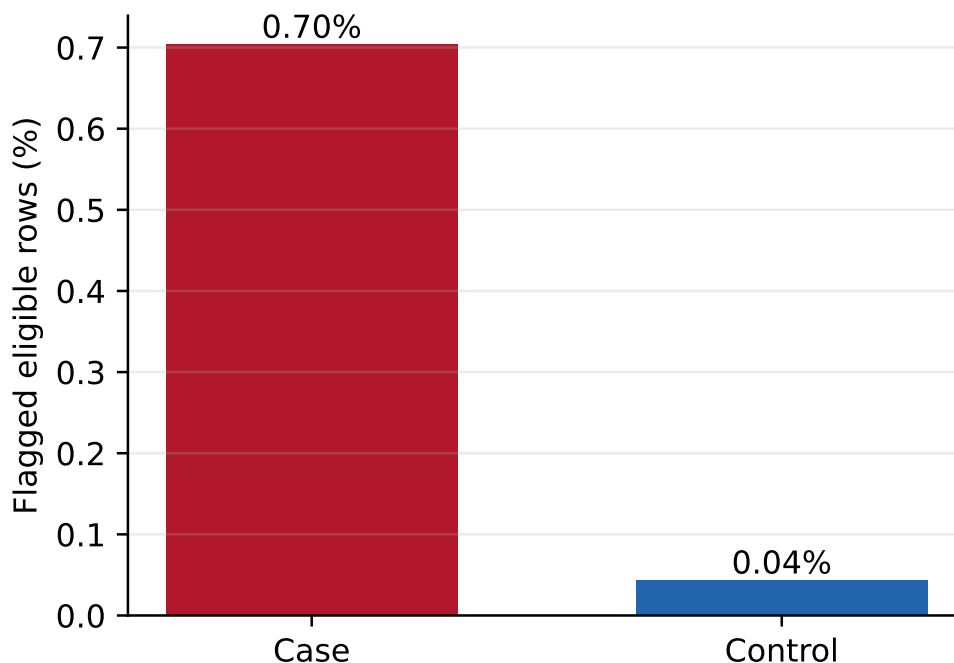


Figure 3: Share of eligible author-subjects flagged in each group under the declared screen. Values and supporting counts are generated from the same final flag used in all downstream analyses.

8.4 Fractional group mixing

93.91% of matched citation weight stayed within the same group (label-swap $p < 0.001$).

Table 10: Defined reciprocal-clique pattern and randomized comparison.

Primary rule	All groups	Recip. groups	Random mean	Random p	Mean density	Density p	Recip. mean	Recip. p	Case share	Label- swap p
$k \geq 4, r \geq 0.50$	50	17	0.73	0.002	0.825	0.469	0.641	0.122	91.6%	0.002

A candidate group is a maximal group in which every pair has a citation in at least one direction. The primary rule requires at least four members, density at least 0.75, and weighted reciprocity at least 0.50. The randomized graphs keep each author’s number of incoming and outgoing links and keep the same set of edge weights, but rewire the links and reassign those weights. Randomized p -values are the fraction of 500 randomized graphs with a value at least as large as the observed value.

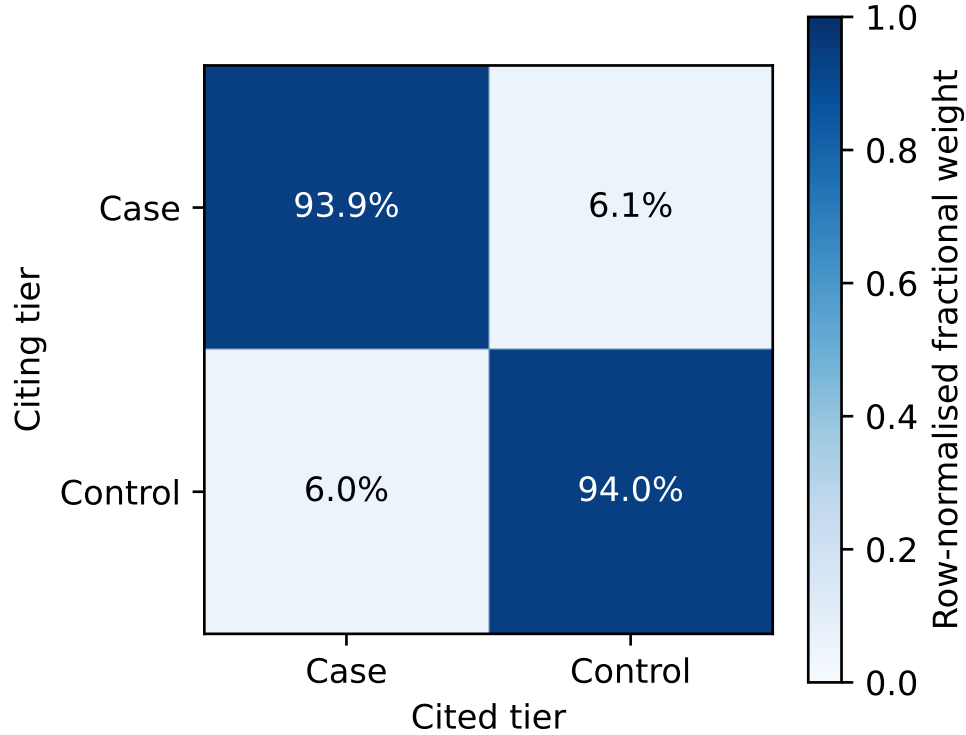


Figure 4: Fractional citation-weight mixing between Case and Control nodes. The comparison distribution is based on within-pair group-label swaps.

8.5 Formal reciprocal-clique structure

The rule found 17 reciprocal cliques among 50 candidate groups. Randomized graphs averaged
295 0.73 reciprocal cliques ($p = 0.0020$). Mean density was 0.825 and mean reciprocity was 0.641.

Table 11: Sensitivity of the clique rule.

Minimum group size	Recip. thresh-old	All groups	Recip. groups	Random mean	Random p	Mean density	Density p	Recip. mean	Recip. p	Case share	Label-swap p
3	0.25	98	73	29.42	0.002	0.859	0.848	0.539	0.002	87.9%	0.002
3	0.50	98	38	5.36	0.002	0.878	0.706	0.693	0.040	89.0%	0.002
3	0.75	98	11	0.54	0.002	0.970	0.378	0.909	0.053	77.8%	0.002
4	0.25	50	35	10.85	0.002	0.806	0.619	0.492	0.002	92.4%	0.002
4	0.50	50	17	0.73	0.002	0.825	0.469	0.641	0.122	91.6%	0.002
4	0.75	50	2	0.01	0.002	1.000	0.125	0.899	0.125	44.4%	0.782
5	0.25	21	11	4.18	0.006	0.852	0.389	0.550	0.004	90.3%	0.002
5	0.50	21	8	0.13	0.002	0.847	0.615	0.649	0.058	87.2%	0.002
5	0.75	21	1	0.00	0.002	1.000	–	1.000	–	0.0%	1.000

The directed density threshold stays at 0.75. The reciprocity threshold is the minimum share of two-way citation weight. Each row uses the same subject-specific candidate groups, and each randomized run scores those same groups.

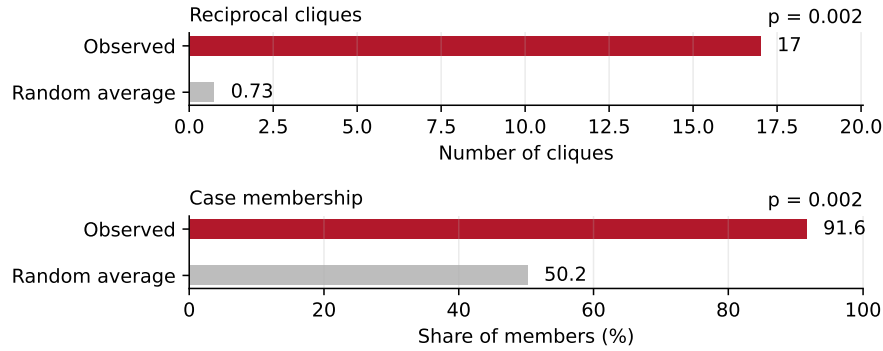


Figure 5: Observed clique results compared with random averages. The top row shows the number of reciprocal cliques; the bottom row shows the percentage of clique members who are Cases. Red bars are observed values, gray bars are random averages, and each row reports its empirical p -value.

Here, a clique is a group of at least four matched authors in which every pair is connected by a citation in at least one direction. We compare these groups with randomized networks that keep each author’s number of incoming and outgoing links and keep the same set of edge weights, but rewire the links and reassign those weights. We also swap Case and Control labels within each matched pair. A small p -value means that few randomized comparisons 300 reached the observed value. These comparisons test whether the pattern is unusual and whether it is concentrated in Cases; they do not say why it occurred.

The clearest differences are the number of reciprocal cliques and the share of their members who are Cases. Mean directed density was 0.825 ($p = 0.4685$), and mean weighted reciprocity was 0.641 ($p = 0.1220$); neither measure was unusual in the randomized networks. The signal 305

is therefore in how often the defined pattern occurs and who belongs to it, not in every observed clique being denser or more reciprocal.

9 Discussion

This study compares two journal-impact groups after matching authors within subject and
310 on recent citation productivity. The result is an association within matched pairs. It is not an estimate of what would happen if an author changed venues, because venue choice, collaboration, topic, and citation practice can affect one another.

The main result is consistent across the four primary measures: Cases had higher average values for citations to previous collaborators, reciprocal citation, local neighbor connections,
315 and concentration of outgoing citations. Three median differences were zero because many pairs had the same zero value. Among non-tied pairs, Case values were more often higher, so the result is a shift across the sample, not a larger value for every Case author. This pattern complements the author-level concentration measures of Evdaimon et al., while measuring the distribution of references sent by an author rather than the number of authors needed to
320 account for received citations (Evdaimon, Ioannidis, et al., 2024).

The secondary results add detail. Cases had less same-journal referencing and larger year-to-year increases to one recipient. These measures describe different parts of citation behavior from the journal-level citation stacking studied by Heneberg (Heneberg, 2016). The author-level screen also showed a 16.14-fold difference in the share of flagged rows, and
325 93.91% of matched citation weight stayed within the same journal-impact group. These are descriptive results that show how the groups differ; they do not identify a cause.

The clique analysis makes the main pattern more specific. We defined a clique as a group of at least four matched authors in which every pair is connected by a citation in at least one direction. We found 17 reciprocal cliques among 50 candidate cliques, compared
330 with 0.73 on average in randomized graphs ($p = 0.002$). Cases made up 91.6% of clique

memberships, compared with about one half after swapping labels within matched pairs ($p = 0.002$). The average density and reciprocity inside the cliques were not themselves unusual in the randomized graphs. Thus the evidence concerns how often the defined pattern occurs and which group its members belong to. It supports the initial directional hypothesis as a qualified structural association, while leaving the reasons for the pattern open. 335

The screen and clique rule are measurement tools for follow-up. They summarize citation links and cannot determine whether a citation was appropriate, coordinated, or intentional. Narrow specialties, stable research teams, and institutional or career-stage differences can also create cohesive citation groups.

10 Limitations 340

The estimates apply to the fixed 2020–2024 snapshot and the author–subject observations that meet the stated inclusion criteria. The graph metrics therefore describe the defined analytic population and graph construction.

The five broad subject categories cannot capture fine topical proximity. The journal classifier was not independently validated against blinded human coding, and the publication-345 weighted Eigenfactor cutoffs remain sensitive to the observed journal network. The 70% portfolio rule, deterministic ORCID sample, and positive- h_5 matching pool define the study population. Matching on subject-specific h_5 improves comparability within that pool but does not balance geography, language, institution, team structure, career age, or unmeasured research topic. 350

Graph metrics depend on population boundaries. Ego-layer measures include identified recipients outside the matched cohort, whereas reciprocity, clustering, balance, and mixing are restricted to matched nodes. Results from the latter should not be interpreted as full-network properties. The 2020–2024 citation window limits observed citation behavior, while historical work and author metadata in the snapshot can left-censor or misdate prior collaborations.355

The year dimension of the edge data supports coauthor timing and the scalar surge-share outcome, but is not analyzed as a group-specific trajectory or group-by-year interaction; we therefore make no claim that a Case–Control gap widens over time.

The anomaly thresholds and Isolation Forest define an operational screen, not a validated
360 diagnostic. Percentile, seed, and leave-one-feature-out sensitivity quantify some model dependence but do not supply ground-truth labels or make the overlapping restriction independent. Finally, structural metadata cannot reveal the textual function, scientific relevance, or motivation of a citation (J. Liu et al., 2024).

11 Conclusion

365 In the matched comparison, Cases showed more cohesive and concentrated citation patterns than Controls. Their average differences were positive on all four primary measures: 0.045 for citations to previous collaborators, 0.129 for reciprocal citation, 0.026 for local neighbor connections, and 0.097 for outgoing citation concentration. All four comparisons remained significant after adjustment for the four primary tests.

370 The defined clique analysis found 17 reciprocal cliques among 50 candidate groups, compared with 0.73 reciprocal cliques on average in randomized graphs ($p = 0.0020$). Cases supplied 91.6% of clique memberships, compared with about one half after within-pair label swaps ($p = 0.0020$). Density and reciprocity inside the cliques were not individually different from the randomized graphs. Taken together, the results support the initial directional
375 hypothesis as a qualified structural association between journal-impact group and citation-network organization. They provide a clear pattern for follow-up without assigning a cause or intent.

12 Author Contributions

Conceptualization: P.-A.S., D.S.; Methodology: P.-A.S.; Software: P.-A.S., G.A.; Data curation: P.-A.S., G.A.; Validation: P.-A.S., G.A., D.S.; Formal analysis: P.-A.S., G.A.; 380 Visualization: P.-A.S.; Writing, original draft: G.A.; Writing, review and editing: P.-A.S., G.A., D.S.; Supervision: D.S.; Project administration: D.S.

13 Competing Interests

The authors declare no competing interests.

14 Funding Information

385

This research did not receive a specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

15 Data and Code Availability

The revised software, aggregate replication materials, and retained-cohort database are archived as Zenodo version 3.0.0 (Spanakis et al., 2026). The archive contains the subject-390 keyed workflow, tests, manuscript sources, aggregate results, figures, the ORCID-linked retained cohort, and the matching covariate used by the canonical analysis.

References

Aksnes, D. W. (2003). A macro study of self-citation. *Scientometrics*, 56(2), 235–246. <https://doi.org/10.1023/A:1021919228368>

395

- Baccini, A., De Nicolao, G., & Petrovich, E. (2019). Citation gaming induced by bibliometric evaluation: A country-level comparative analysis. *PLOS ONE*, 14(9), e0221212. <https://doi.org/10.1371/journal.pone.0221212>
- Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and
400 powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57(1), 289–300.
- Bergstrom, C. T., West, J. D., & Wiseman, M. A. (2008). The eigenfactor metrics: A network approach to assessing scholarly journals. *Journal of Neuroscience*, 28(45), 11433–11434. <https://doi.org/10.1523/JNEUROSCI.0003-08.2008>
- 405 Biagioli, M., Kenney, M., Martin, B. R., & Walsh, J. P. (2019). Academic misconduct, misrepresentation and gaming: A reassessment. *Research Policy*, 48(2), 401–413. <https://doi.org/10.1016/j.respol.2018.10.025>
- Biagioli, M., & Lippman, A. (Eds.). (2020). *Gaming the metrics: Misconduct and manipulation in academic research*. MIT Press. <https://doi.org/10.7551/mitpress/11087.001.0001>
- 410 Efron, B., & Tibshirani, R. J. (1993). *An introduction to the bootstrap*. Chapman & Hall.
- Evdaïmon, I., Ioannidis, J. P. A., et al. (2024). Metrics to detect small-scale and large-scale citation orchestration. *arXiv Cornell University*. <https://doi.org/10.48550/arXiv.2406.19219>
- Garfield, E. (2006). The history and meaning of the journal impact factor. *Journal of the
415 American Medical Association*, 295(1), 90–93.
- Goodhart, C. A. E. (1984). Problems of monetary management: The UK experience [Originally published in 1975 by the Reserve Bank of Australia under “Papers in monetary economics”]. In *Monetary theory and practice* (pp. 91–121). Macmillan Education UK. https://doi.org/10.1007/978-1-349-17295-5_4
- 420 Heneberg, P. (2016). From excessive journal self-cites to citation stacking: Analysis of journal self-citation kinetics in search for journals, which boost their scientometric indicators. *PLoS ONE*, 11(4), e0153730. <https://doi.org/10.1371/journal.pone.0153730>

- Hicks, D., Wouters, P., Waltman, L., de Rijcke, S., & Rafols, I. (2015). Bibliometrics: The Leiden Manifesto for research metrics. *Nature*, 520(7548), 429–431. <https://doi.org/10.1038/520429a> 425
- Hirsch, J. E. (2005). An index to quantify an individual’s scientific research output. *Proceedings of the National Academy of Sciences*, 102(46), 16569–16572. <https://doi.org/10.1073/pnas.0507655102>
- Hirschman, A. O. (1964). The paternity of an index. *The American Economic Review*, 54(5), 761–762. 430
- Kojaku, S., Livan, G., & Masuda, N. (2021). Detecting anomalous citation groups in journal networks. *Scientific Reports*, 11(1), 14524. <https://doi.org/10.1038/s41598-021-93572-3>
- Liu, F. T., Ting, K. M., & Zhou, Z.-H. (2008). Isolation forest. *Proceedings of the 8th IEEE International Conference on Data Mining*, 413–422. 435
- Liu, J., Bai, X., Wang, M., Tuarob, S., & Xia, F. (2024). Anomalous citations detection in academic networks. *Artificial Intelligence Review*, 57(4), 103. <https://doi.org/10.1007/s10462-023-10655-5>
- LLM Guidelines Consortium. (2024). *LLM Guidelines: Transparency in LLM-assisted research*. Retrieved September 17, 2025, from <https://llm-guidelines.org/> 440
- Newman, M. E. J. (2002). Assortative mixing in networks. *Physical Review Letters*, 89(20), 208701. <https://doi.org/10.1103/PhysRevLett.89.208701>
- Spanakis, P.-A., Alexandrou, G., & Spinellis, D. (2026). *Replication package for: Citation-network cohesion across journal-impact strata* (Version v3.0.0). Zenodo. <https://doi.org/10.5281/zenodo.21720914> 445
- Spinellis, D. (2023). Open reproducible scientometric research with Alexandria3k. *PLoS ONE*, 18(11), e0294946. <https://doi.org/10.1371/journal.pone.0294946>

Van Noorden, R., & Chawla, D. S. (2019). Hundreds of extreme self-citing scientists revealed in new database. *Nature*, 572(7771), 578–579. <https://doi.org/10.1038/d41586-019-02>

479-7

Wilcoxon, F. (1945). Individual comparisons by ranking methods. *Biometrics Bulletin*, 1(6), 80–83.

Wilhite, A. W., & Fong, E. A. (2012). Coercive citation in academic publishing. *Science*, 335(6068), 542–543. <https://doi.org/10.1126/science.1212540>

A Upstream Venue-Score Specification

For subject s , let $Z_{jk}^{(s)}$ count citations from citing journal j to cited journal k . The diagonal is removed, and each non-dangling citing-journal column of H_s is

$$(H_s)_{kj} = \frac{Z_{jk}^{(s)}}{\sum_{\ell} Z_{j\ell}^{(s)}}. \quad (9)$$

Let a_s be the normalized journal article-count vector and d_s indicate dangling columns. With damping $\alpha = 0.85$, the subject-specific influence vector is obtained by

$$\pi_s = \alpha H_s \pi_s + [\alpha d_s^\top \pi_s + (1 - \alpha)] a_s. \quad (10)$$

The received-influence vector $H_s \pi_s$ is normalized to sum to 100 within subject. Thus citation mass flows from citing to cited journals, journal self-citations are excluded, and dangling mass and restart mass follow article counts. These fixed journal-level scores are upstream inputs to the author matching and are not author-network outcomes.

B Matching and Sensitivity Specifications

Cases and Controls require at least three eligible works and a 70% portfolio share in the relevant journal group. The retained matching pool additionally requires a materialized positive h_5 value; zero- h_5 profiles are not candidates. Within that pool, candidate eligibility uses the inclusive hard caliper $|h_{5,\text{Case}} - h_{5,\text{Control}}| \leq 3$. Greedy selection proceeds without replacement within subject in the total order subject, ascending absolute h_5 distance, Case ORCID, and Control ORCID. The retained cohort must pass the same subject-keyed positive- h_5 join and caliper checks before reuse. Exact- h_5 pairs form a sensitivity subset rather than a replacement cohort.

The paired bootstrap resamples pair identifiers with replacement. The sign-flip test acts on complete-pair differences for one metric at a time. Multiple-testing adjustment is not pooled across primary and secondary families. The same-year coauthor sensitivity changes only $c_{ij} < t$ to $c_{ij} \leq t$; all other definitions and pair membership remain fixed.

C Anomaly-Screen Specification

Controls define subject-specific robust centers, scales, detector fits, score cutoffs, and univariate cohesion cutoffs. Cases do not affect those reference quantities. The canonical feature output retains the detector score, cohesion-restriction indicator, citation eligibility, and final flag for each (ORCID, s , tier) row. The downstream mixing routine consumes that final flag rather than recomputing or substituting a label.

The primary screen uses seed 42 and the 99th Control percentiles. The threshold analysis repeats the complete fit at the 97.5th, 99th, and 99.5th percentiles for ten fixed seeds. A row with an unavailable required detector input is not zero-imputed; it receives no detector score, cannot be flagged, and is identified by a separate detector-completeness field. The overlap table is restricted to rows on which both criteria are defined.

Table 12: Screen stability when one input is removed.

Detector feature omitted	Flags	Case flags	Control flags	Enrichment	Overlap with main screen
Self-citation rate	53	49 (0.70%)	4 (0.04%)	16.14	0.963
Coauthor-citation rate	50	46 (0.66%)	4 (0.04%)	15.15	0.907
Reciprocity	52	48 (0.69%)	4 (0.04%)	15.81	0.981
Local clustering	55	51 (0.73%)	4 (0.04%)	16.80	0.964
Outgoing HHI	51	47 (0.68%)	4 (0.04%)	15.48	0.962

Each row removes one screen input. The cutoff and random seed stay fixed, and the Case and Control percentages use the same eligible rows. Overlap is the Jaccard share of flagged rows retained by both screens.

D Reproducibility Checks

The rebuild checks the following manuscript-facing invariants:

- 490 • (ORCID, s , tier) is unique in the feature output;
- retained pairs have complete subject-keyed h_5 records, satisfy the inclusive three-point caliper, and do not reuse an author–subject key;
- every analytic edge has non-null endpoints and is unique on (s, i, j, t) ;
- each per-metric complete-pair count is no greater than 9,431;
- 495 • every observed surge share lies in $[0, 1]$;
- the same final flag supplies every flag-dependent enrichment and figure artifact; and
- reported numerical values enter through generated LaTeX artifacts rather than hand-copied prose.

SQL fixtures separately test fractional weight, null endpoints, work-reference deduplication, 500 subject retention, and prior-coauthor timing. Python unit tests cover matching eligibility and deterministic order, retained-cohort validation, metric formulas, paired inference, subject-keyed joins, detector overlap, feature ablation, weighted mixing, and flag reuse.

E Journal Subject Classification

The retained journal mapping was produced with OpenAI `gpt-4.1` (snapshot `gpt-4.1-2025-04-14`) at zero temperature from journal names and representative work titles and abstracts.505 Prompts and cached outputs are part of the replication materials. Because the mapping has no independent human-validation sample and no open-weight baseline, classification error remains a limitation rather than a measured source of uncertainty (LLM Guidelines Consortium, 2024).